Alessandro Caula

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Profile

Versatile Software Developer and Bioinformatician with expertise in C#, Python, and JavaScript, and a passion for creating efficient, data-driven solutions for scientific applications. Skilled in software engineering, algorithm design, data processing, and visualization, I build robust user-friendly applications. Committed to continuous improvement, my adaptability, creativity, and clear communication have helped me develop tools that enhance research and support users.

Technical Skills

Programming Languages: C#, Python, JavaScript, TypeScript, Bash, HTML, CSS
Frameworks & Libraries: .NET, React, Node.js, Tailwind CSS, Biopython, SciPy
Machine Learning & Data Science: NumPy, Pandas, Scikit-learn, Keras, Nextflow
Visualization: Matplotlib, Seaborn, Plotly, DevExpress, LightningChart
OS & Tools: Windows, Linux, macOS, Git, VS Code, Google Colab, AWS, MongoDB, Docker

Experience

Bioinformatics Software Developer, *3Brain AG – Genova, ITALY* May 2022 – Sept 2025

Designed and developed high-performance scientific software for real-time electrophysiological data acquisition, analysis and visualization in *BrainWave*, a **C#/.NET** application used by neuroscience researchers. Key contributions:

- Developed and optimized algorithms for signal processing, frequency analysis (3× faster) and trajectory detection.
- Efficiently processed large datasets (50–80 GB/recording) through scalable pipelines, ensuring high performance.
- Engineered modular, object-oriented components, ensuring scalability and maintainability.
- Built interactive and intuitive GUIs and charts for real-time data exploration and visualization.
- Implemented and integrated statistical and machine learning algorithms into core pipelines.
- Worked in a large, fast-moving codebase, collaborating across teams to deliver maintainable, robust software.
- Improved performance and robustness through rigorous debugging and architectural refactoring.

Data Scientist Intern, *University of Lisbon – Lisbon, PORTUGAL* Apr 2021 – Feb 2022

Developed and implemented supervised and unsupervised Machine Learning pipelines for omics data analysis using **Python**, aiming to identify molecular subtypes of Alzheimer's disease ([*GitHub*] [*Paper*]).

- Preprocessed and cleaned data, performed feature selection and modeling, and evaluated results with key metrics.
- Created visualizations and summaries of model outputs, clearly communicating biological insights to researchers.
- Applied a range of ML methods including Non-negative Matrix Factorization, Support Vector Machines, Naïve Bayes, *k*-Nearest Neighbor, Decision Trees, Logistic Regression, Random Forests, and XGBoost.

Laboratory technician, *IHMA Sarl – Monthey, SWITZERLAND* Feb 2019 – Jul 2019

In vitro studies of new and existing antimicrobials for Phase II, III, and IV programs.

- Performed sample processing, organism identification (MALDI-TOF), and antimicrobial susceptibility testing.
- Prepared and validated testing panels, maintained quality control, and supported data recording.

Education

University of Bologna, *M.Sc. in Bioinformatics* Oct 2019 – Feb 2022

Gained a strong foundation in computational biology, algorithms implementation, data science, and machine learning.

- **Key coursework:** Algorithms and Data Structures, Python and Bash programming, Basic and advanced Machine Learning, Systems and in Silico Biology, Big Data Processing Infrastructures, DNA/RNA dynamics

University of Turin, *B.Sc. in Biology* Sept 2015 – Oct 2018

- **Key coursework:** Cellular and Molecular Biology, Biochemistry, Human and Plant Physiology, Microbiology, Chemistry, Public Health, Statistics and Scientific Analysis.

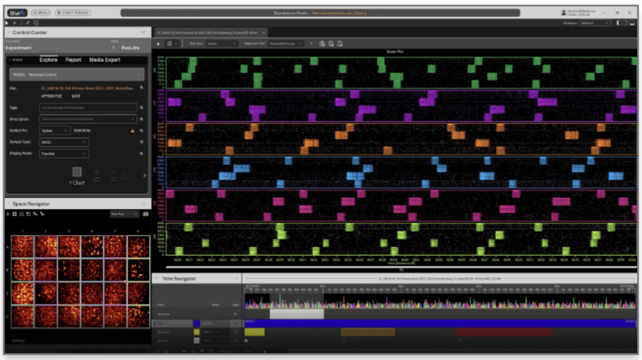
Languages

Italian - Native **English** - Fluent (professional proficiency) **French** - Beginner

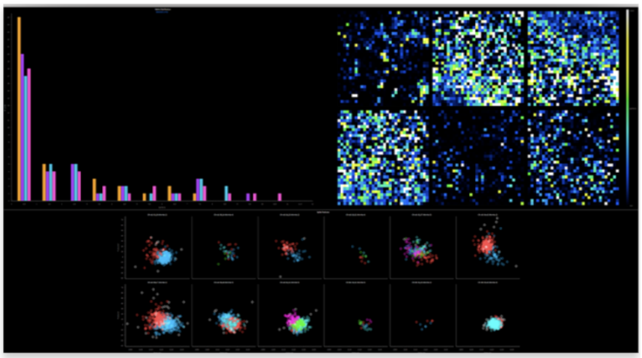
Project Highlights

BrainWave - 3Brain AG

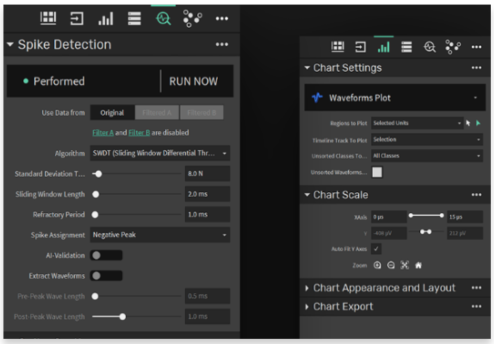
High-performance **C#/.NET** software for real-time electrophysiology data acquisition, analysis, and visualization.



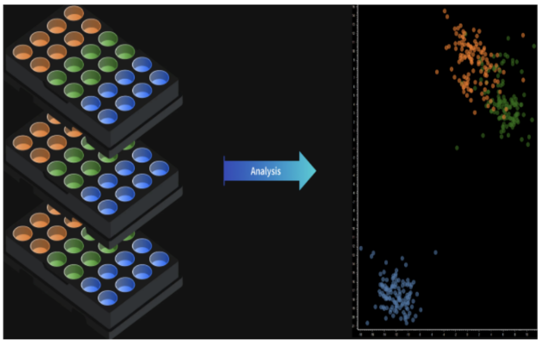
Modular, high-performance software with interactive GUI components.



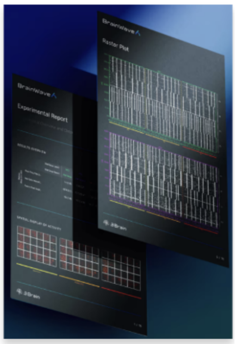
Interactive and dynamic visualizations of electrophysiological analysis.



Fully customizable and intuitive graphical interface.



ML and clustering on multi-channel recordings.



Automated Report.

Bioinformatic pipeline - Protein Loop Reconstruction

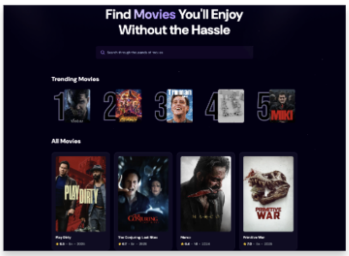
Python pipeline for automatic identification and modeling of missing internal loops in protein structures:

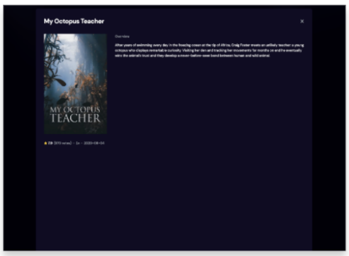
- Generated multiple loop conformations with the CCD algorithm, and evaluated for steric clashes, Ramachandran outliers, and anchor RMSD.
- Designed modular, extendable pipeline for structural bioinformatics applications.

Web Development

Practiced front-end and full-stack web development using **React.js** and modern libraries, applying best practices for modular and maintainable code.

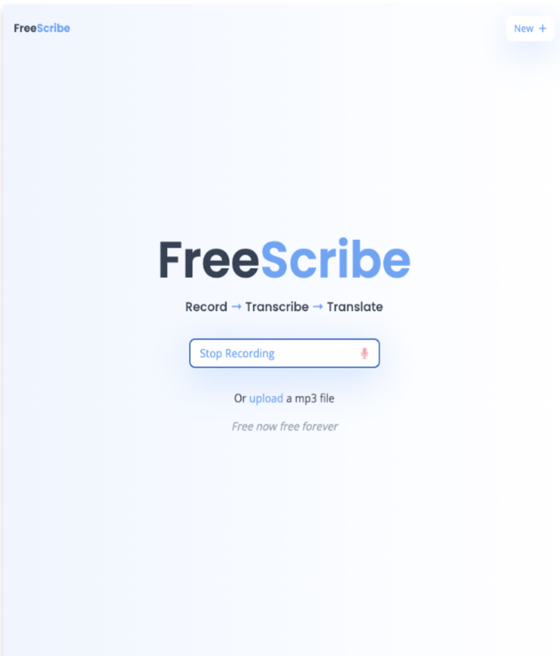
- Worked with APIs, backend communication, and real-time processing (including Web Workers for AI tasks).
- Strengthened skills in modular frontend architecture and user-friendly interface design.





React app for browsing and discovering movies via external API.

Modern landing page built with React, Tailwind, and GSAP.



React app for audio-to-text translation using OpenAI.

3Brain AG
Huobstrasse 16
8808 Pfäffikon SZ
Switzerland

Genova, 04 September 2025

Letter of Reference

3Brain AG is a Swiss deep tech company working on cell-electronic interfaces that link biological networks to computers via custom-made semiconductor microchips.

We hereby certify that Mr Alessandro Caula, born on 12/06/1996, was employed at 3Brain AG (3Brain) from 02/05/2022 until 01/09/2025, on indefinite term and full-time contract. He held the position of Bioinformatics Software Developer, within the Software Department.

Mr Caula has fulfilled the following responsibilities:

- Developing data processing tools and algorithms, with a strong emphasis on software architecture and quality, exploring data storage and optimization strategies with a proactive approach and problem-solving attitude.
- Developing algorithms and strategies for statistical analysis of high-volume biological datasets, and techniques for presenting and visualizing results on charts and diagrams, through a user-friendly graphical interface.
- Contribution to scientific and technical activities related to the Company's products, including testing/optimization of new algorithms, analysing customer data, and supporting the scientific publication process.
- Maintaining of existing and released software – active help and support in planning, testing, identification and resolution of bugs.

Mr Caula played a key role, and his expertise in bioinformatics, development of data processing and analysis algorithms, and data visualization strategies, was instrumental in helping us process and interpret complex biological datasets with high-level performances and precision. During his time at 3Brain, he consistently proved to be a quick and talented learner. He demonstrated excellent skills to apply combined creative and logical approach to programming tasks, that helped the overall process of developing innovative tools and high-level optimization strategies. His code was clean and reliable, reflecting a thoughtful approach to software engineering.

In addition to his technical skills and professional competencies, he exhibited a deep understanding of biological data and experimental workflows, which contributed to facilitate the high-quality communication across different teams.

His dynamic, positive, friendly and caring personality is a very important contribution for a collaborative and constructive team environment, and to the overall workplace wellness.

Mr Caula left on 01/09/2025, free of any commitment, except as regards professional secrecy.

We can strongly recommend Mr Caula to any future employer, and we wish him all the best for the continuation of his professional career.

A Godjo

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